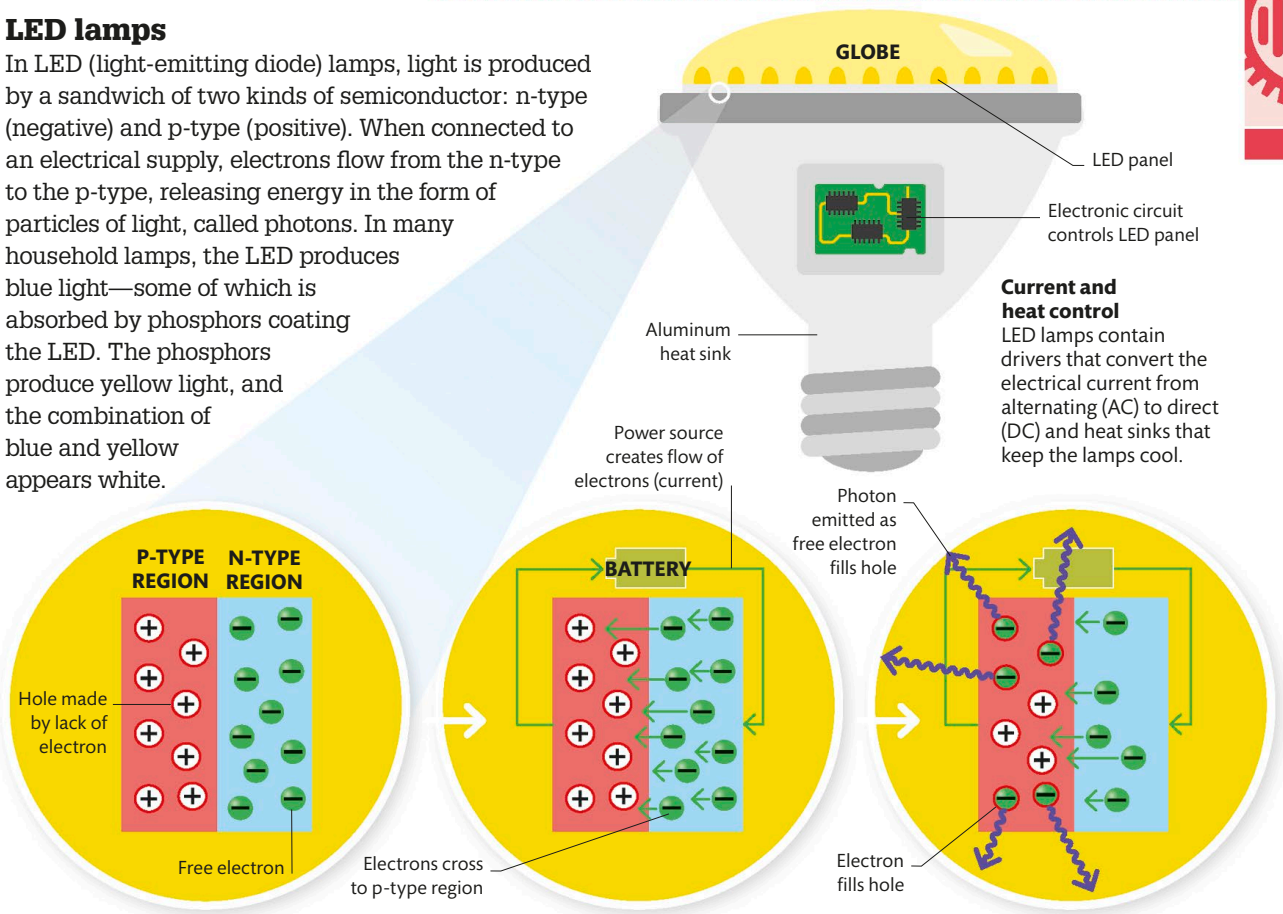
LED lamps

In LED (light-emitting diode) lamps, light is produced by a sandwich of two kinds of semiconductor: n-type (negative) and p-type (positive). When connected to an electrical supply, electrons flow from the n-type to the p-type, releasing energy in the form of particles of light, called photons. In many household lamps, the LED produces blue light—some of which is absorbed by phosphors coating the LED. The phosphors produce yellow light, and the combination of blue and yellow appears white.



1 Semiconductors
The semiconductors in most LEDs are compounds of the element gallium. Adding traces of other elements creates n-type and p-type regions, in which there are too many or too few electrons.

2 Flow of electrons
Connecting an electrical supply across a junction between the regions pushes electrons from the n-type region into the p-type region, where they fill the holes that result from the lack of electrons.

3 Photons
When an electron fills a hole, it falls into a lower energy level in an atom of gallium, and its energy loss releases a photon. An LED produces billions or trillions of photons every second.

LIGHT SOURCES (EQUAL BRIGHTNESS)



CFL
Power consumption 18W
Average life span
8,000 hours



LED
Power consumption 9W
Average life span
25,000 hours



INCANDESCENT
Power consumption 60W
Average life span
1,200 hours

INCANDESCENT BULBS

Up to the end of the 20th century, the most common form of domestic electric lighting was the incandescent light bulb. Inside, a thin, coiled length of tungsten wire called a filament becomes white hot when an electric current flows through it. The wire does not burn, because the bulb is filled with inert gases rather than air. However, its incandescence produces light.

